MIL-D-36786B
9 December 1985
SUPERSEDING
MIL-D-36786A
14 March 1974

MILITARY SPECIFICATION

DRILL, PIN HOLE, DENTAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers spiral (twist) drills suitable for drilling pin holes in the dentin of a tooth for the support of the pin retention restorative techniques.

1.2 Classification. Drills covered by this specification shall be of the following classes, sizes and operative lengths as specified (see 6.2).

Classes.

Class AL. For use with latch angle handpiece.
Class FG. For use with friction grip angle handpiece.

Sizes (Diameter, inches)

Size 1.	0.021
Size 2.	0.024
Size 3.	0.027
Size 4.	0.017
Size 5.	0.0135

Operative lengths (millimeters)

2
5
1.3
1.5

APPLICABLE DOCUMENTS

2.1 Government documents.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding Paperboard.
PPP-B-601	Boxes, Wood, Cleated-plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-P-116	Methods of Preservation.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-151	Metal, Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packaging of.

DRAWINGS

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

22160	Drill, Pin Hole, Dental
23318	Drill, Pin Hole, Dental, Contra-Angle

(Copies of specifications, standards, and drawings, required by contractors in connection with specific acquisition functions, should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 92 Standard Test Method for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries on other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Drills of one-piece construction shall be fabricated from tungsten-vanadium tool steel. Drills of two-piece construction shall have the head fabricated from tungsten-vanadium tool steel welded and securely held inside an aluminum alloy shank.

3.2 Style, design and dimensions. The spiral drill shall be suitable for drilling pin holes in the dentin of a tooth for the support of the pin retention restorative techniques, and shall conform to the requirements as specified herein for the class, size and operative lenth, as applicable. Class AL drill shall be suitable for use in latch contra-angle handpieces. The drill shall be of one-piece construction or two-piece construction. The drill shall be furnished in the classes, sizes and operative lengths as specified in Table I.

TABLE I. Drills.

Class	Size	Operative length millimeters (mm)
AL	1	2
AL	1	5
AL	2	5
AL	3	2
AL	3	5
AL	4	1
AL	5	1.3
FG	1	2
FG	3	2

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The style, design and dimensions shall conform to Defense Personnel Support Center (DPSC) drawings 22160 and 23318 for class, size, and operative length, as specified. As an option, Class AL, size 1, Operative length (mm) 2, shall have a groove approximately 1/16 inch from the distal end of the shank. The groove shall measure .003 ± .001 inch.

3.3 Hardness. The hardness of the drill head (working end) shall be 700 to 840 on the Vickers scale (Diamond pyramid hardness), or shall have a Knoop hardness number of 737 to 860, when tested as specified in 4.4.2.

3.4 Runout. The total indicated runout shall not exceed 0.0010 inch, when measured as specified in 4.4.3.

3.5 Finish. The operative end of the drill shall have a fine machine finish and shall be properly sharpened. The shank and neck shall have a smooth finish.

3.6 Identification markings. Latch type, angle handpiece (Class AL), of one-piece construction shall be permanently and legibly marked with the name or registered trademark of the contractor or manufacturer. In addition, drills of foreign manufacture shall be marked with the name of the country of origin.

3.6.1 Color coding. The shank of each drill shall be color coded as follows:

Color	Head diameter (inches)
Gold	.027
Black	.024
Silver	.021
Red	.017
Pink	.0135

3.7 Workmanship. The drill shall be free from defects which detract from its appearance or impair its serviceability, proper functioning or intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physicaland chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in Tables II and IV (see 4.2.1 and 4.3) and Table III (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptance quality levels (AQLs) shall be as indicated in Table II. Unit of product for examination purposes shall be the unit of issue.

TABLE II. Examination criteria.

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S-1	1.5
Minor A	II	2.5
Minor B	S-1	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table III, except that the acceptance number shall be zero for all sample sizes. The unit of product for all tests except chemical analysis shall be the unit of issue.

TABLE III. Test Criteria.

Characteristic	Requirement	Test procedure	Inspection level
Chemical analysis	3.1	4.4.1	See 4.2.3
Hardness	3.3	4.4.2	S-1
Runout	3.4	4.4.3	S-4

4.2.3 Sampling of raw materials. A sufficient quantity of each lot of raw material used in the fabrication of the drills shall be selected for chemical analysis.

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4.2.4 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. Drills shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in Table IV.

TABLE IV. Classification of Defects.

Categories	Defects
Critical	None defined.
Major A:	
101	Class, size and length of drills not as specified.
102	Finish (fine machined and properly sharpened) of operative end of drills not as specified.
103	Latch portion of Class AL drills not cleanly machined and not free of burrs.
Major B:	
151	Dimensions of latch end of shank, Class AL drills, not as specified.
152	Diameter of shank not as specified.
153	Twist end of diameter not as specified.
154	Twist end length not as specified.
Minor A:	
201	Packaging, packing, and marking not as specified.
202	Number and size of drills in unit package not as specified.
203	Shanks of drills not smoothly finished.
204	Shanks not color coded.
Minor B:	
251	Dimensions of drills, other than Major B, not as specified.

4.3.1 Dimensions. Examination of dimensions shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with them.

4.3.2 Microscopic examination. Examination of drill defects not readily perceptible to the naked eye, shall be made using a microscope at 30 power (30X) magnification.

4.4 Tests

4.4.1 Chemical analysis of raw material. In the absence of certificates of quality, the chemical composition of the steel used in the manufacture of the drills shall be determined using Method 111 or 112 of FED-STD-151.

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4.4.2 Hardness. Testing for hardness shall be conducted in accordance with American Society for Testing and Materials (ASTM) Method E-92 for Vickers hardness. Knoop hardness shall be determined on a Tukon tester under a 500 gram load, using a rhomus-shaped, diamond indenting tool with a longitudinal angle of 172.5 degrees and a transverse angle of 130 degrees.

4.4.3 Runout. The total indicated runout of the drill head shall be measured to the nearest 0.0005 inch. The drill shall be held in a precision chuck and rotated slowly. An optical comparator, toolmaker's microscope, or other suitable equipment of equal accuracy shall be employed. The difference between the maximum and minimum readings for one full rotation of the drill head shall be taken as the total indicated runout.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Drills shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Drills shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit package. Six drills of one class, size, and operative length only, shall be packaged in a plastic holder, so designed that each drill is contained in an individual cell. The plastic holder shall be designed with a clear plastic cover in order to prevent spilling of contents under normal handling. Each unit shall be further protected by being packaged in accordance with MIL-P-116, method IC-1.

5.1.1.5 Intermediate package. Twelve units containing drills of one class, size, and operative length only, shall be packed in a box conforming to PPP-B-566 or PPP-B-676; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for drills shall conform to MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 Level A. Fifty intermediate packages containing drills of one class, size, and operative length only, shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-601, overseas type; PPP-B-621, class 2: or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed in accordance with PPP-B-636.

Note: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B. Fifty intermediate packages containing drills of one class, size, and operative length only, shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of boxes shall be as specified in the applicable box specifiation.

5.2.3 Level C. The level C packing for drills shall conform to MIL-STD-794 requirements for this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Levels A and B. Drills, packed as specified in 5.2.1 and 5.2.2, shall be unitized on pallets conforming to MIL-STD-147.

5.2.4.2 Level C. Drills, packed as specified in 5.2.3, shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B, and C. Each unit pack, intermediate pack, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

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5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

5.5 Regulatory requirements.

5.5.1 Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U. S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

5.5.2 Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

6. NOTES

6.1 Intended Use. The spiral drills covered by this specification are intended for drilling pin holes in the dentin of a tooth for the support of pin retention restorative techniques.

6.2 Ordering data.

6.2.1 Acqusition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class, size and operative length desired (see 1.2 and 3.2, Table I).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.4 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

6.5 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

Stock Number	Item Identification
6520-00-017-4930	DRILL, PIN HOLE, DENTAL, LATCH ANGLE HANDPIECE, 0.021 inch diameter by 5 mm Operative Length

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6520-00-149-0565	DRILL, PIN HOLE, DENTAL, FRICTION GRIP ANGLE HANDPIECE, 0.027 inch diameter by 2 mm Operative Length
6520-00-149-0571	DRILL, PIN HOLE, DENTAL, FRICTION GRIP ANGLE HANDPIECE, 0.021 inch diameter by 2 mm Operative Length
6520-00-420-0510	DRILL, PIN HOLE, DENTAL, LATCH ANGLE HANDPIECE, 0.027 inch diameter by 2 mm Operative Length
6520-00-420-0511	DRILL, PIN HOLE, DENTAL, LATCH ANGLE HANDPIECE, 0.021 inch diameter by 2 mm Operative Length
6520-00-935-1097	DRILL, PIN HOLE, DENTAL, LATCH ANGLE HANDPIECE, 0.024 inch diameter by 5 mm Operative Length
6520-00-935-1098	DRILL, PIN HOLE, DENTAL, LATCH ANGLE HANDPIECE, 0.027 inch diameter by 5 mm Operative Length
6520-01-036-2699	DRILL, PIN HOLE, DENTAL, LATCH ANGLE HANDPIECE, 0.017 inch diameter by 1.5 mm Operative Length
6520-01-095-3707	DRILL, PIN HOLE, DENTAL, LATCH ANGLE HANDPIECE, 0.0135 inch diameter by 1.3 mm Operative Length
6520-01-119-6397	DRILL, PIN HOLE, DENTAL, LATCH ANGLE HANDPIECE, 0.017 inch diameter by 1.5 mm Operative Length

Military Interests:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review activities:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:

DoD - MB

Agent:

DLA - DM

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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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